Paleomagnetism of the Monteregian Hills: New Results

A. LAROCHELLE

Geological Survey of Canada, Ottawa, Ontaria, Canada

Paleomagnetic directions were obtained for seventy-four stably and homogeneously magnetized cores drilled at sixteen sites on four of a group of eight basic intrusive plugs located in southeast Quebec and known as the Monteregian Hills. A statistical analysis of these data shows that the mean directions (regardless of polarity) of magnetization of the four sampled bodies were not significantly distinct after the cores were 'cleaned' in an alternating field of 250 oe. The circle of confidence about the mean of the sixteen site means (each given unit weight) has a radius, α_{95}, of 3.8°. Because of the high precision of these data it is proposed that they supersede a set of data obtained from the same series of rocks and published five years ago. The seventy-four directions yielded a paleomagnetic pole at 171.2°W, 69.9°N. This pole position is in remarkably good agreement with the poles derived from six different areas in North America and at present regarded as reliable estimates for the Cretaceous period. The high coherency of the Cretaceous poles so far estimated from North American rocks strongly supports the view that the earth's magnetic field was dipolar at that time.

INTRODUCTION

In a previous study [*Larochelle*, 1962] the mean of a set of paleomagnetic directions was used to estimate the age of a group of eight basic intrusive plugs located in southeastern Quebec and known as the Monteregian Hills. As it turned out, the validity of the estimated Cretaceous age was later overwhemingly confirmed by several K/Ar and Rb/Sr age determinations [*Fairbairn et al.*, 1963] and, accordingly, authors have referred to the pole position derived from the Monteregian Hills as a reliable North American Cretaceous pole. Nevertheless, it must be recognized that the precision of these previously reported data is relatively low and that the apparently accepted pole position calculated on their basis may be in error by several degrees. It may be suspected that the relatively high scattering of these paleomagnetic directions can largely be attributed to the numerous sources of error inherent in the sampling, measuring, and cleaning techniques and in instrumentation available to the writer at the time this study was made (1957–1958). It thus recently appeared worth while to resample some of the hills in order to check this possibility and, if necessary, to revise the earlier data. In this paper, the results obtained after re-examining four of the hills are reported and analyzed.

SAMPLING AND MEASUREMENTS

Ninety-one cores were drilled with a portable diamond drill [*Gaucher and Meilleur*, 1965] from eighteen sites distributed among St. Bruno, Shefford, Brome, and Johnson mountains (Figure 1). Depending on the size of the individual intrusive bodies, the sampling sites were spaced between a half and one and a half kilometers apart, so as to obtain a proportional coverage of each sampled body.

An average of five cores per site were drilled (generally at least 3 meters apart) and oriented in situ with a magnetic compass. When sun conditions permitted (for more than half of the cores), the cores were also oriented in situ with a solar compass. From a comparison of the two orientations, it was found that an error of as much as 20° could result in determining the paleomagnetic declination of a core by relying solely on the magnetic compass orientation, although the error was generally much less, i.e., of the order of 3° to 5°.

From each core, 3.2 cm in diameter, two 2.8-cm-long cylinders were cut and labeled according to their position relative to the bottom of the core.

The direction and intensity of natural remanent magnetization (NRM) of each cylinder were determined with a biastatic magnetometer [*Larochelle and Christie*, 1967], and the mean

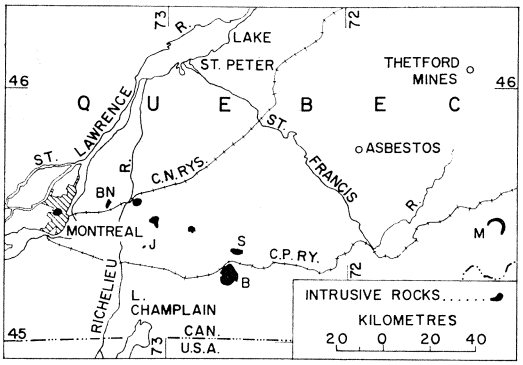


Fig. 1. Map showing distribution of Monteregian Hills. BN, St. Bruno; S, Shefford; B, Brome; J, Johnson; M, Megantic.

NRM direction (direction cosines of resultant and intensity of each pair of cylinders were computed and taken as those of the core they represented.

Stability Test

The bottom cylinders of the second cores drilled at each site were used for a series of stability tests with the apparatus described by *Larochelle and Black* [1965]. The peak value of the demagnetizing field was first set at 50 oe, and the residual magnetization in each of the eighteen test cylinders was measured after a first treatment in this field. The operation was repeated with the demagnetizing peak field

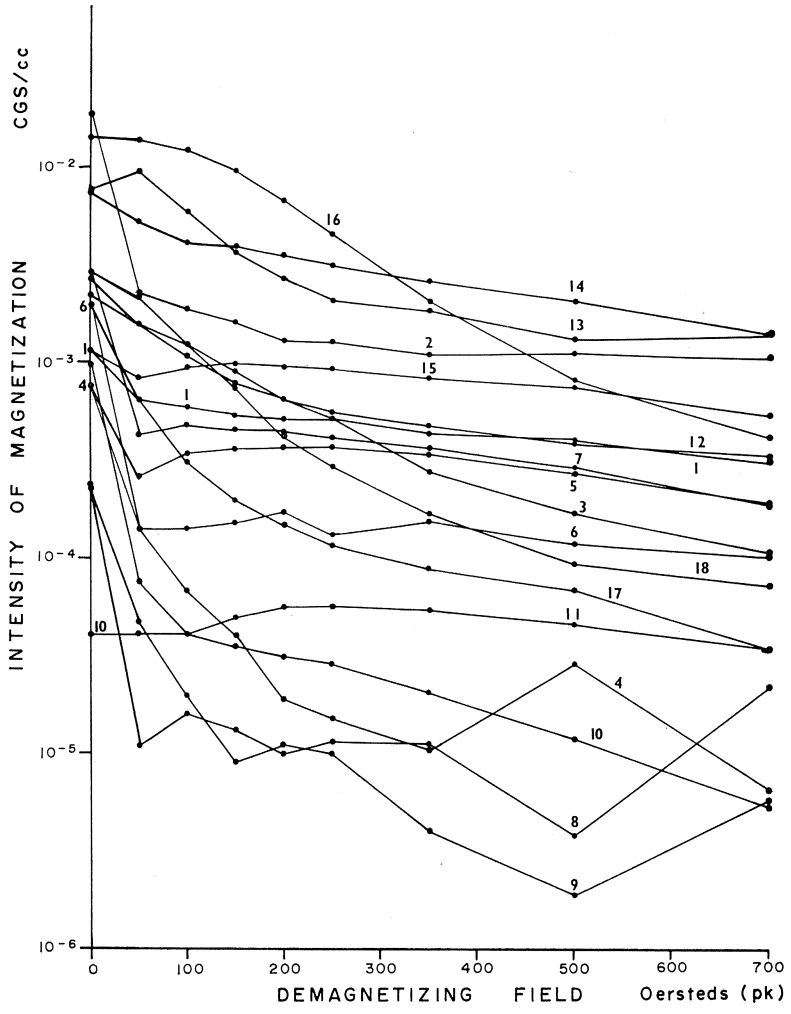
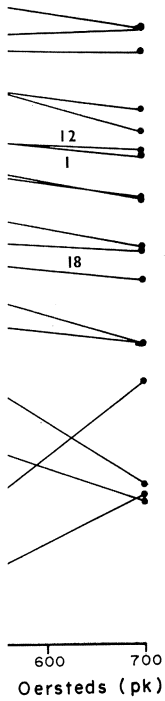


Fig. 2. Demagnetization curves of eighteen test cylinders. Numbers refer to the site of their origin.

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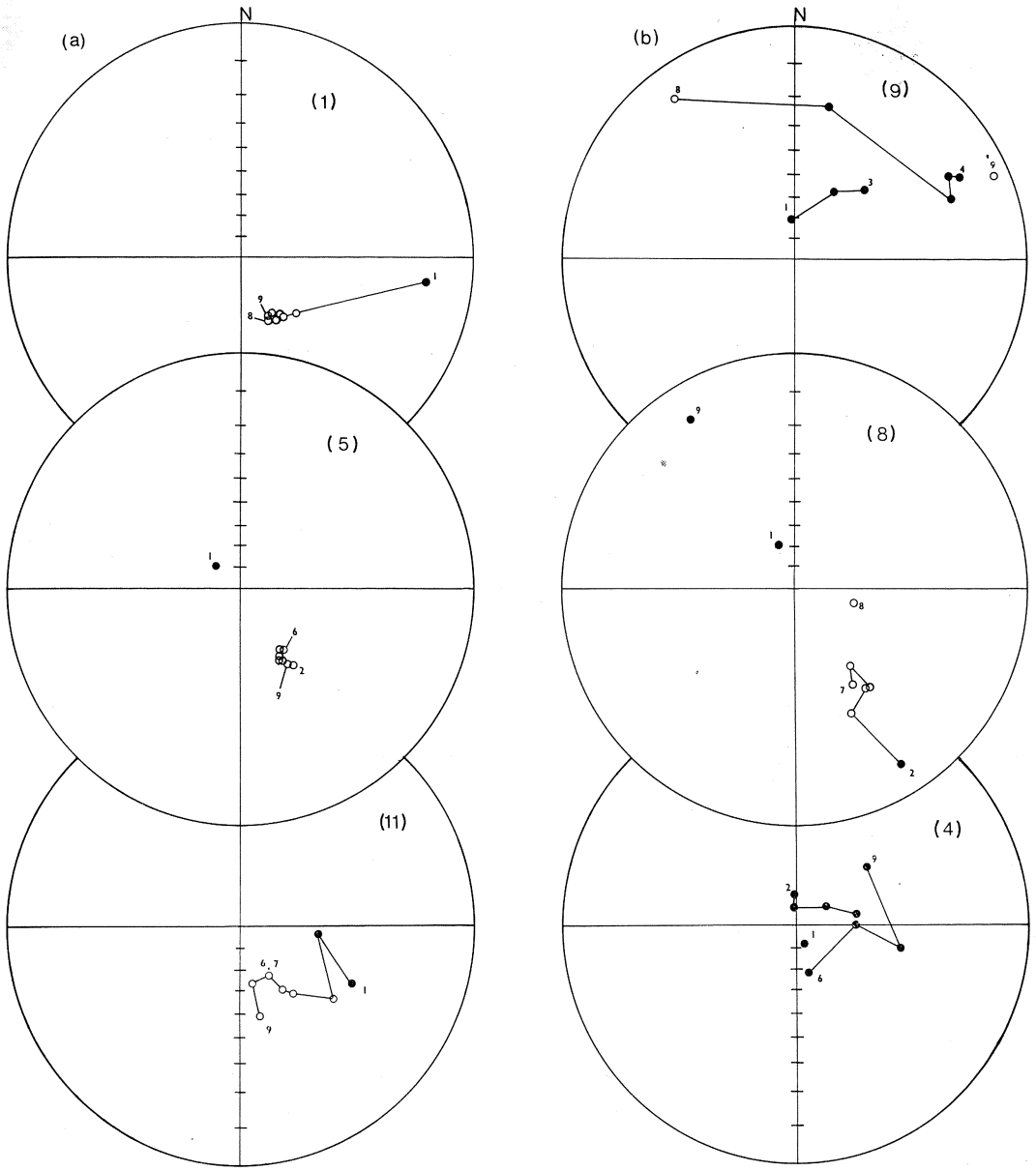


Fig. 3. Stereographic projection of magnetization vectors of typical (*b*) unstable (*a*) stable test specimens following demagnetizing treatments. Dots (circles) indicate downward (upward) pointing vectors; numbers refer to treatment in 0 (1), 50 (2), . . . , 700 (9) oe. Numbers in parentheses refer to test specimen numbers; lines join successive treatments.

successively set at 100, 150, 200, 250, 350, 500, and 700 oe.

The demagnetization curves (Figure 2) show that after the 200-oe treatment the intensities generally decreased slowly as the field peak was increased. Notable exceptions to this rule are the intensities of the cylinders representative of sites 4, 8, and 9, which decreased sharply after the first treatments and varied erratically throughout the rest of the series of tests. The directions of magnetization of these test cylinders were also found to have

changed appreciably after each treatment (Figure 3*b*), in contrast with the other cylinders (Figure 3*a*).

The intensity of several of the test specimens (5, 6, 7, 11, and 15) reached a minimum after either the first or the second treatment. This is easily explained by the probably incomplete elimination after these treatments of a strong component of viscous magnetization opposing more or less the more stable but smaller thermoremanent component. The bulk of the viscous component in other specimens was less important, and its partial elimination was displayed on the demagnetization curves by a maximum.

Magnetic Cleaning

An inspection of Figure 2 suggests that a demagnetizing field of 200 oe was sufficient to remove the viscous magnetization from most of the specimens dealt with, but the remaining 164 cylinders in the collection were nevertheless 'cleaned' in a demagnetizing field of 250 oe. Since the directions of magnetization were pointing downward for the majority of the samples collected at sites 16, 17, and 18, these cores were submitted to an additional treatment in a peak demagnetizing field of 500 oe. No significant changes were observed in the directions of magnetization of the individual cylinders from these three sites, thus confirming an observation already made during the stability tests.

The angle θ between the directions of magnetization of each pair of cleaned cylinders was computed, and it was found that this angle exceeded 15° for fourteen cores. Ten of these cores were collected at sites 4, 8, or 9 of which the representative cylinders had given signs of instability during the stability tests. It is noted further that some difficulty had been encountered in finding suitable drilling spots at sites 4 and 8, owing to deep weathering of the surface rock. The bottom specimens of about half of the fourteen inhomogeneously magnetized cores were magnetized within a few degrees from the corresponding site mean direction, whereas the magnetization of the top cylinders sometimes diverged by as much as 100° from that direction. As it appeared probable that the stable magnetization of these cylinders was made up largely of a secondary component related to recent chemical changes in the rock, their directions were not thought to be parallel to that of the earth's field at the time the Monteregian Hills were formed, and they were not considered further.

One core from site 18 was found to have a mean NRM intensity about 100 times as large as that of the other cores at that site. A similar discrepancy prevailed even after the 250- and 500-oe cleaning treatments. A cylinder of this core and one of another core from site 18 were heated to 700°C and allowed to cool in the earth's field. As the intensity of the two artificially magnetized cylinders were rigorously equal, it was concluded that most of the stable NRM of the first one was probably acquired as a result of lightning. The data obtained from this core will not be considered further in the rest of this study.

The mean directions of magnetization of the remaining seventy-six cores before and after magnetic cleaning were computed. Site mean directions and site angular standard deviations

$$\delta_s = [2(N - R)/(N - 1)]^{1/2}$$

are listed in Table 1. In the last column of this table the site average values of θ for the homogeneously magnetized cores are given. These values indicate that the probable measurement error was less than 2° in most cases.

Discussion of Results

An inspection of Table 1 illustrates clearly that the NRM of the rocks dealt with bears an important component of unstable viscous magnetization, which is often directed against the more stable thermoremanent component. This is suggested by the drastic reduction in the angular standard deviation at most of the sites after cleaning and by the coherency of site means among the reversely polarized ones. The consistency of the data obtained from the normally polarized sites of Mount Johnson is also excellent, and the magnetization vectors there are almost diametrically opposed to those of the other intrusives. This appears to be a case of simple reversal of the earth's field, as discussed in detail in the earlier study mentioned above.

The only two homogeneously magnetized cores from site 4 might suggest at first that mixed polarities are to be expected on Shefford

TABLE 1. Summary of Magnetization Directions

Site	Int.	N	n	NRM D, deg	NRM I, deg	NRM δ_s, deg	After Cleaning in 250-oe Demagnetizing Field D, deg	After Cleaning in 250-oe Demagnetizing Field I, deg	After Cleaning in 250-oe Demagnetizing Field δ_s, deg	After Cleaning in 250-oe Demagnetizing Field θ_{avg}, deg
1	BN	5	4	103.9	−7.9	40.5	150.6	−58.6	3.3	3.6
2*	BN	5	5	316.2	14.1	45.0	144.2	−57.9	9.5	3.8
3*	BN	5	5	178.5	−20.2	47.7	165.2	−46.7	10.1	6.2
4*	SH	5	2	347.9	68.7	5.0	14.8	66.8	5.8	4.0
5	SH	6	6	224.1	−24.5	69.2	156.4	−62.9	4.9	2.4
6*	SH	6	6	254.1	57.9	67.5	170.5	−50.3	10.6	4.8
7*	SH	6	6	251.2	59.2	45.0	143.9	−62.1	6.2	3.5
8*	BR	5	2	99.0	87.8	14.4	144.4	−45.5	9.6	6.8
9*	BR	4	0	...	...	...	...	...	...	...
10	BR	4	3	199.4	54.3	73.6	161.7	−50.7	10.4	5.7
11	BR	5	5	88.1	64.3	47.7	145.9	−60.6	4.6	3.2
12	BR	4	4	16.8	56.3	28.9	152.8	−51.5	7.2	6.1
13	BR	6	6	180.7	−68.3	32.4	155.0	−60.7	8.7	4.5
14	BR	4	3	176.8	−43.9	23.8	156.8	−57.2	2.5	8.7
15	BR	5	5	26.5	37.2	30.0	147.9	−65.4	5.8	4.2
16	J	6	6	336.6	70.5	12.8	325.2	64.8	2.1	2.4
17	J	5	4	111.6	66.8	67.5	344.1	68.4	8.2	7.1
18	J	5	4	335.3	−2.9	73.6	328.9	54.3	17.2	4.4
All		91	74†	...	...	...	154.0	−58.5	10.6	4.5

Notes

N, number of cores collected; n, number of observations used in computing D, I, δ_s, and θ_{avg}.
δ_s, angular standard deviation.
θ_{avg}, average angular deviation between the magnetization direction of pairs of cylinders.
Int., intrusive body on which sites are located: BN, St.-Bruno; SH, Shefford; BR; Brome; J, Johnson.
* Sites for which only magnetic compass was used for orientation.
† Data from site 4 excluded and polarity of data from sites 16, 17, and 18 reversed.

Mountain, unlike on the other sampled intrusives. Their thermoremanent magnetization is questionable, however, not only because deep weathering was encountered at this site but also because their NRM is so consistently aligned along the present earth's field. It appears that these cores acquired their present stable magnetization predominantly as new minerals were formed during the weathering process. Accordingly, they were considered as unreliable samples, and their remanent magnetization directions were not included in the following statistical analysis.

The small angular standard deviation at most sites almost precludes the likelihood of important orientation errors during the sampling. The dispersions could easily be explained by local anomalies in the ancient geomagnetic field and, to some extent, by its secular variations. Small components of chemical magnetization may also be present in some of the cores.

Statistical Analysis

The Mount Johnson cores being magnetized almost diametrically opposite to the other cores, their polarity was reversed for the sake of the following statistical analysis. The seventy-four magnetization directions considered reliable were grouped by sites, and, for each intrusive sampled, the between-site (δ_b^2) and the within-site (δ_w^2) angular variances were computed. The between-site angular variance is defined by

$$\delta_b^2 = 2(N - R')/(B - 1)$$

where N and B are the total number of cores and sites, respectively, and R' is the projection (on the mean direction of the N vectors) of B vectors whose directions are the same as the site means and whose lengths are the numbers of cores from the respective sities. The within-site variance is given by

TABLE 2. Angular Variance-Ratio Analysis (within Intrusives)

Intrusive	B	N	D, deg	I, deg	δ_w^2, radians^2	δ_b^2, radians^2	δ_b^2/δ_w^2	$F_{2(B-1),2(N-B),0.05}$
St. Bruno	3	14	154.6	−54.5	0.0224	0.1276	5.7	2.82
Shefford	3	18	158.4	−58.9	0.0178	0.1825	10.2	2.69
Brome	7	28	152.2	−57.8	0.0149	0.0625	4.2	2.00
Johnson*	3	14	150.9	−63.1	0.0317	0.0908	2.86	2.82
4 intrusives	(4)	74	154.0	−58.5	(0.0334)	(0.0500)	1.5	2.15
4 intrusives	(4)	(16)	153.7	−57.6	(0.0225)	(0.0152)	0.7	2.51
4 intrusives	16	74	154.0	−58.5	0.0202	0.0922	4.6	1.57

Notes

B, number of sites (intrusives); N, number of cores (site means).
δ_w^2 and δ_b^2 within-site (intrusive) and between-site (intrusive) angular variances.
D and I, declination and inclination of resultants.
* Polarity of data reversed.

$$\delta_w^2 = 2(\sum^{B} R_i - R)/(N - B)$$

where R_j is the resultant length at the jth site and R is the length of the N vectors resultant. The variance ratio test postulates that, if δ_b^2/δ_w^2 is greater than the statistic $F_{2(B-1),2(N-B),0.05}$, there are good reasons to assume that the site mean directions are significantly distinct. The application of the test to the data obtained from the three reversely polarized hills show that the site means are significantly distinct within each of these intrusives (Table 2). The fact that the test was not conclusive with the Mount Johnson data is compatible with the close spacing of the sites on this relatively small body.

The seventy-four directions were next grouped by intrusives, regardless of sites. It was indicated on the basis of the test that the intrusive means were not significantly distinct. When the site means were given unit weights and were grouped by intrusives, it was suggested even more strongly that the intrusive means were not significantly distinct. Finally, when the seventy-four directions were grouped by sites, regardless of intrusives, the site means were found to be significantly distinct.

The most conservative estimates of the standard angular deviation of the mean of the N vectors is thus given by

$$\delta_{\bar{R}}' = \delta_b/\sqrt{N}$$

and the confidence limit at the .95 probability level by

$$\alpha_{95}' = \sqrt{3}\,\delta_{\bar{R}}' = 3.5°$$

Although it cannot be stated on the basis of the present data by how many million years Mount Johnson differs in age from any of the other three sampled intrusives, it seems clear that it is not rigorously contemporaneous with any one of them, because of the difference in polarities. An attempt was made to evaluate the possibility that any two of the reversely

TABLE 3. Angular Variance Ratio Analysis (between Intrusives)

Intrusive Pair	N	δ_w^2, radians^2	δ_b^2, radians^2	δ_b^2/δ_w^2	$F_{2,2(N-2),0.05}$
St. Bruno-Shefford	6	0.0283	0.0102	0.4	4.46
St. Bruno-Brome	10	0.0207	0.0040	0.2	3.63
Shefford-Brome	10	0.0218	0.0198	0.9	3.63

Notes

N, number of site means.
δ_w^2 and δ_b^2, within-intrusive and between-intrusive angular variances.

polarized hills were intruded contemporaneously. The site means of pairs of intrusives were grouped, and their within-intrusive and between-intrusive angular variances were computed. In each case the ratio δ_b^2/δ_w^2 (Table 3) is smaller than the corresponding statistic F, indicating that the three intrusives probably cooled simultaneously. This naturally confirms the previous result that the intrusive means are not significantly distinct. Furthermore, the small angular standard deviation (8.1°) of the thirteen site means involved suggests that the cooling of the three intrusives probably took place within a few thousands and perhaps within a few hundred years.

Pole Position

With the assumption of a central dipole field in Cretaceous time, a pole position was calculated on the basis of the above-given mean direction of magnetization of the Monteregian Hills. This pole (171.3°W, 69.9°N) is at the center of an ellipse of confidence of semi-axes 5.0° and 3.7°, respectively. It is in surprisingly good agreement with six others (Table 4) that were derived from widely separated North American Cretaceous formations. This more than confirms the validity of the conclusions arrived at on the basis of earlier work on the age of the Monteregian Hills, the correlation of Mount Megantic with them, and the age of the Isachsen diabase dykes. In fact, the extremely low dispersion of the seven poles listed in Table 4 strongly supports the hypothesis of the dipolar nature of the earth's magnetic field in Cretaceous time. A similar view may also be suggested about the Triassic period and perhaps about the Permo-Carboniferous, as illustrated in Figure 4 by the discrete clustering of reliable poles for these two periods.

Conclusion

That the data reported in the present study should supersede the earlier paleomagnetic results on the Monteregian Hills hardly needs to be defended. The mean direction of magnetization is based on a larger number of independent observations (74 against 49), thus diminishing the effect of any possible gross errors in any individual observation. The standard error of the mean is appreciably smaller in the present data, even when the mean of the sixteen normalized site means is considered. Finally, the chances of gross experimental errors being considerably reduced with the presently used techniques, it would not be realistic to grant both the old and the new sets of data the same weight and to estimate the probable stable direction of magnetization of the Monteregian Hills on their mean.

It may be argued that only four of the seven previously sampled hills were considered in the present study and that further studies on the other hills might yield highly discordant results. Owing to the coherency of the intrusives mean directions of magnetization, this occurrence is unlikely, however, and even the earlier results, despite their high dispersion, also suggest this.

The pole position arrived at in the present study should accordingly supersede the one based on the earlier measurements. When this pole is compared with others derived from far distant Cretaceous rock units in North America, it is found also to be in surprisingly better agreement with them.

TABLE 4. Paleomagnetic Pole Positions Derived from North American Formations of Cretaceous Age

Rock Unit	Locality	N*	Pole Position	Reference
1. Mount Megantic intrusive	SE Quebec	6	171°E, 68°N	*Larochelle* [1961]
2. Isachsen diabase	Canadian Arctic	10	180°, 69°N	*Larochelle et al.* [1965]
3. Granitic plutons	California	14	165°W, 69°N	*Grommé and Merrill* [1965]
4. Mount Ascutney intrusives	SE Vermont	2	173°W, 65°N	*Opdyke and Wensink* [1966]
5. Elkhorn Mtn. volcs	SW Montana	13	171°W, 69°N	*Hanna* [1967]
6. Mesaverde formation	SW Wyoming	4	177°E, 70°N	*Kilbourne* [1967]
7. Monteregian Hills	SE Quebec	16	173°W, 69°N	This paper

* The number of site means on which the derived pole positions are based.

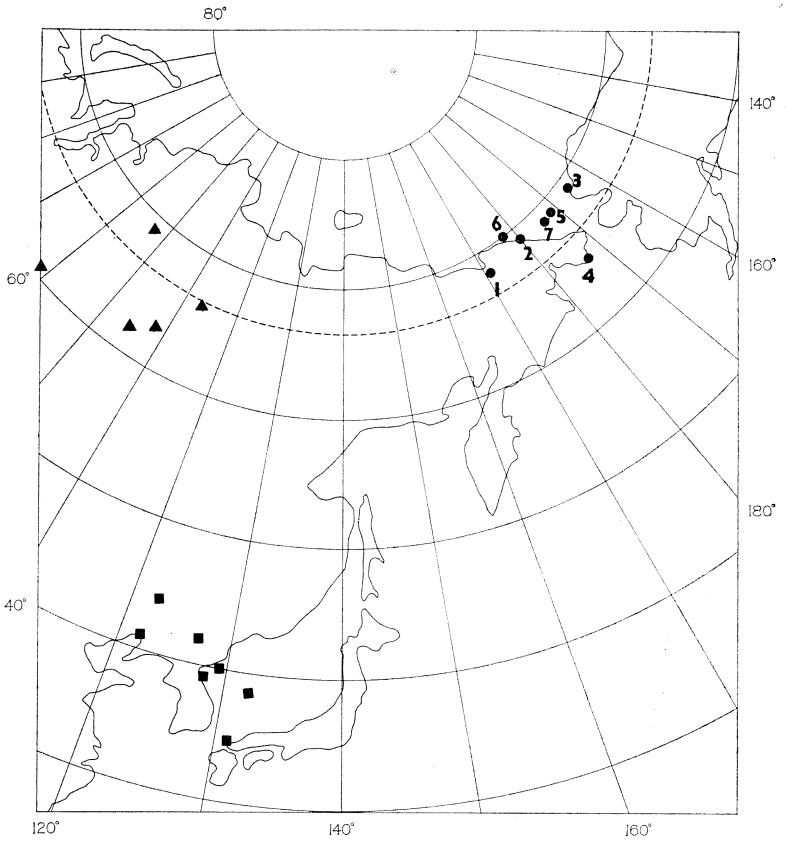


Fig. 4. Geographic distribution of Cretaceous (dots), Triassic (triangles), and Permo-Carboniferous (squares) pole positions derived from North American formations. For co-ordinates of last two sets see *Larochelle* [1967]. Numbers refer to Table 4 entries.

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